CLOSED SUBSPACES OF SCHEMES ARE SCHEMES WITH QUASI-COHERENT IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1979-1995, following the statement at 1960-1977.

Editorial note. The closing 'In particular' sentence of the statement follows from items (1)-(4); the proof covers all four items.

Allowed standard prerequisites (editorial).

- Closed immersions of an affine scheme come from ideals. lemma-closed-immersion-affine-case (card L26394, cited at ~1983). Identifies i^{-1}(U) -> U with Spec(R/I) -> Spec(R).

Source reference. Original lemma 01IH: sources/stacks-schemes/pinned-original-source.tex, source lines 1817–1817. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let $i : Z \to X$ be a closed immersion of locally ringed spaces.*

(1) *The locally ringed space Z is a scheme,*
(2) *the kernel $\mathcal{I}$ of the map $\mathcal{O}_X \to i_*\mathcal{O}_Z$ is a quasi-coherent sheaf of ideals,*
(3) *for any affine open $U = \mathrm{Spec}(R)$ of X the morphism $i^{-1}(U) \to U$ can be identified with $\mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$ for some ideal $I \subset R$, and*
(4) *we have $\mathcal{I}|_U = \widetilde{I}$.*

In particular, any sheaf of ideals locally generated by sections is a quasi-coherent sheaf of ideals (and vice versa), and any closed subspace of X is a scheme.

Proof. Let $i : Z \to X$ be a closed immersion. Let $z \in Z$ be a point. Choose any affine open neighbourhood $i(z) \in U \subset X$. Say $U = \mathrm{Spec}(R)$. By Lemma 01IH we know that $i^{-1}(U) \to U$ can be identified with the morphism of affine schemes $\mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$. First of all this implies that $z \in i^{-1}(U) \subset Z$ is an affine neighbourhood of z. Thus Z is a scheme. Second this implies that $\mathcal{I}|_U$ is $\widetilde{I}$. In other words for every point $x \in i(Z)$ there exists an open neighbourhood such that $\mathcal{I}$ is quasi-coherent in that neighbourhood. Note that $\mathcal{I}|_{X \setminus i(Z)} \cong \mathcal{O}_{X \setminus i(Z)}$. Thus the restriction of the sheaf of ideals is quasi-coherent on $X \setminus i(Z)$ also. We conclude that $\mathcal{I}$ is quasi-coherent. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.